

Bus Arbitration

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- Bus:

A bus is a shared communication link, which uses one set of wires to connect multiple subsystems.

- Buses

Synchronous Bus

Asynchronous Bus

- Synchronous Bus

If a bus is synchronous (e.g. Processor-memory), it includes a clock in the control lines and a fixed protocol for communicating that is relative to the clock.

Synchronous bus

- Synchronous buses have two major disadvantages: -
 - First, every device on the bus must run at the same clock rate.
- Second, because of clock skew problems, synchronous buses cannot be long if they are fast.

Asynchronous bus

- An asynchronous bus is not clocked.
- It can accommodate a wide variety of devices, and the bus can be lengthened without worrying about clock skew or synchronization problems.
- To coordinate the transmission of data between sender and receiver, an asynchronous bus uses a handshaking protocol.

BUS ARBITRATION

-It is selection mechanism required to enable the current master i.e. (Bus Controller) to decide among competing request.

Bus arbitration schemes usually try to balance two factors:

- Bus priority: the highest priority device should be serviced first
- Fairness: Even the lowest priority device should never be completely locked out from the bus.

Types of Arbitration:

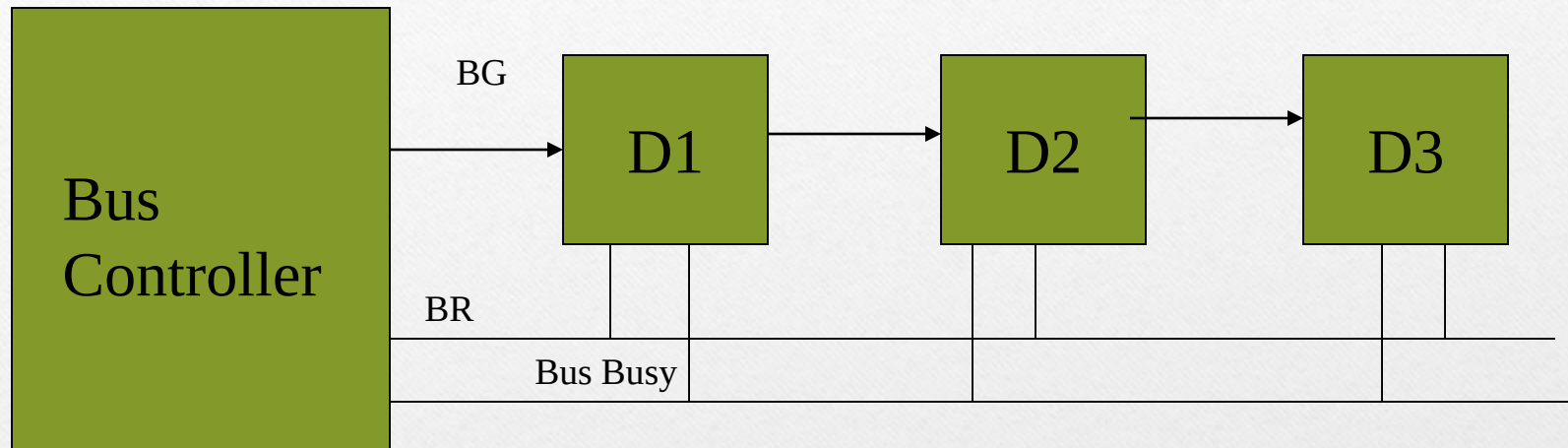
static: (priority fixed)

- Daisy chaining
- Parallel arbitration
- Independent Requesting.

dynamic: (flexible priority)

- polling
- Time slice
- LRU/FIFO
- rotating daisy chain

Daisy chain Arbitration



Advantage: simple

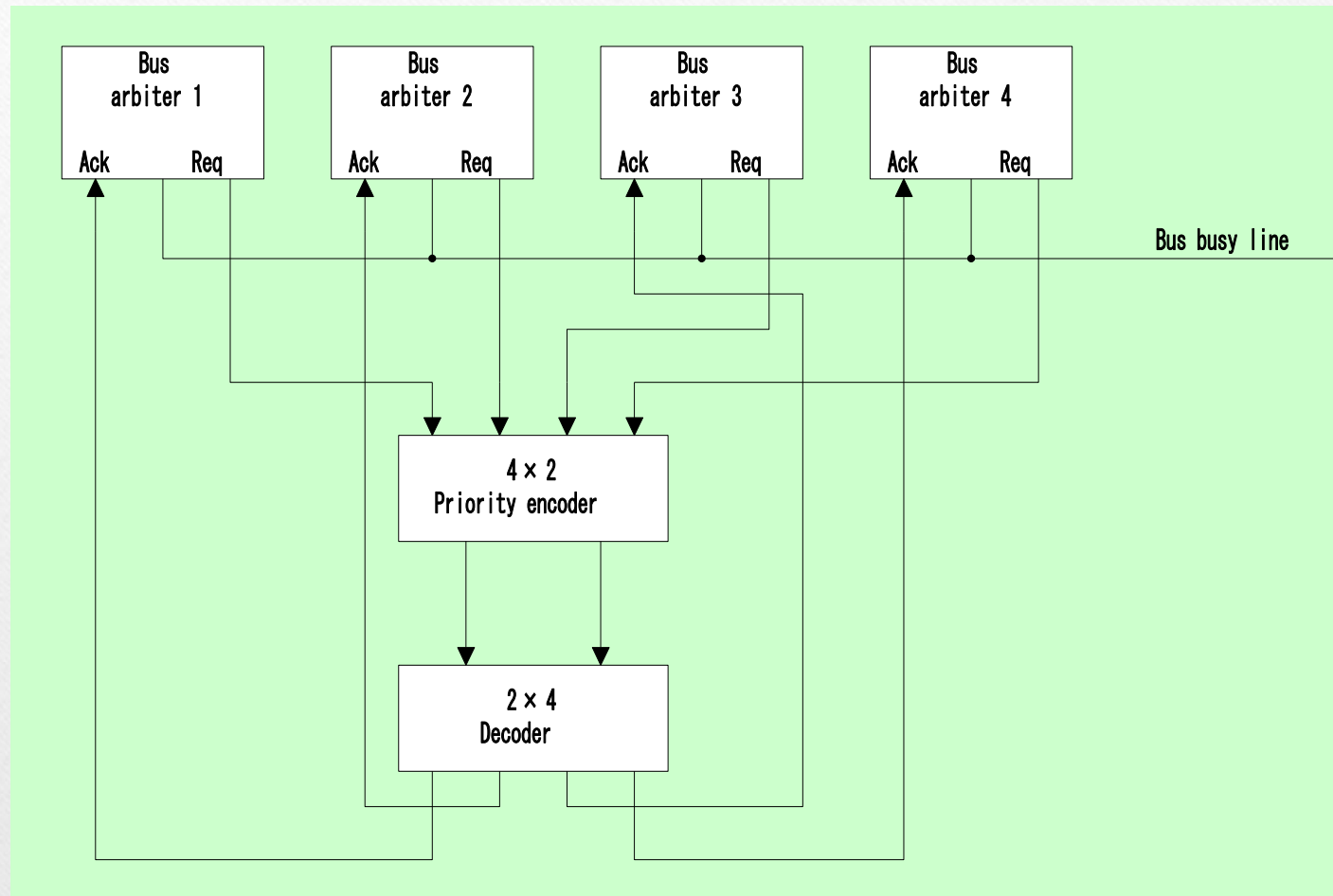
- Disadvantages:

- Cannot assure fairness:

A low-priority device may be locked out indefinitely

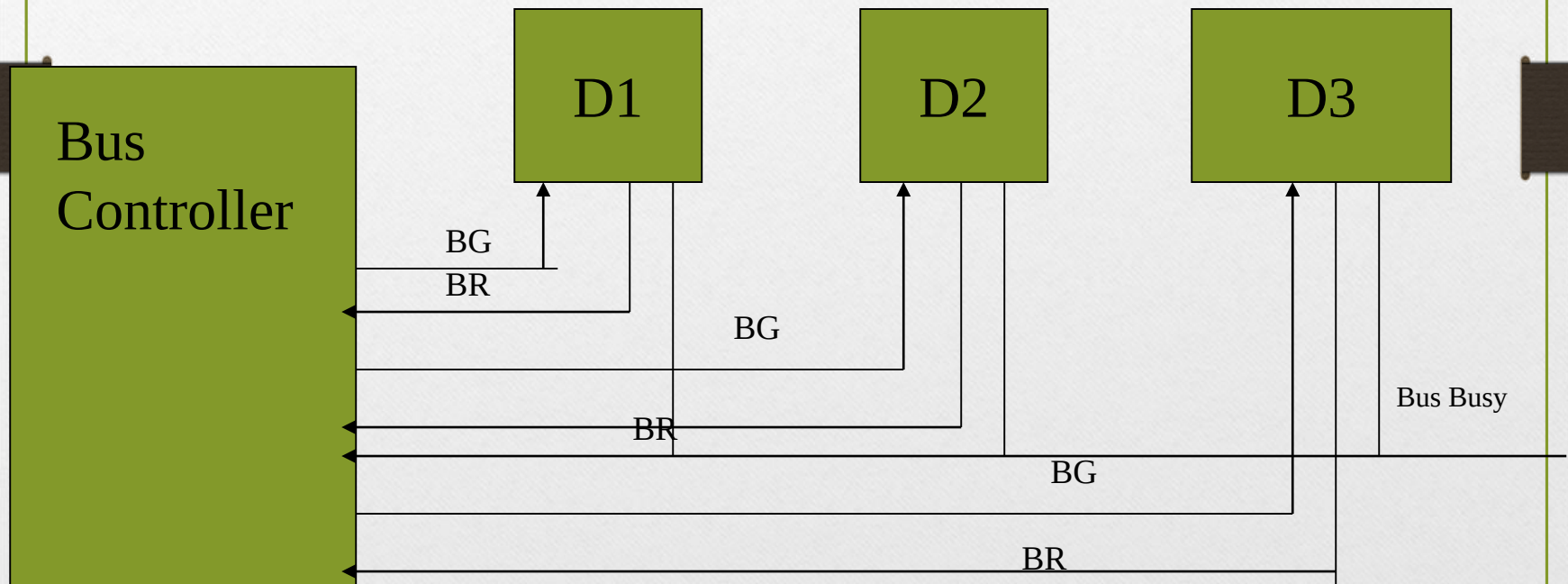
- The use of the daisy chain grant signal also limits the speed of the bus

Parallel Arbitration

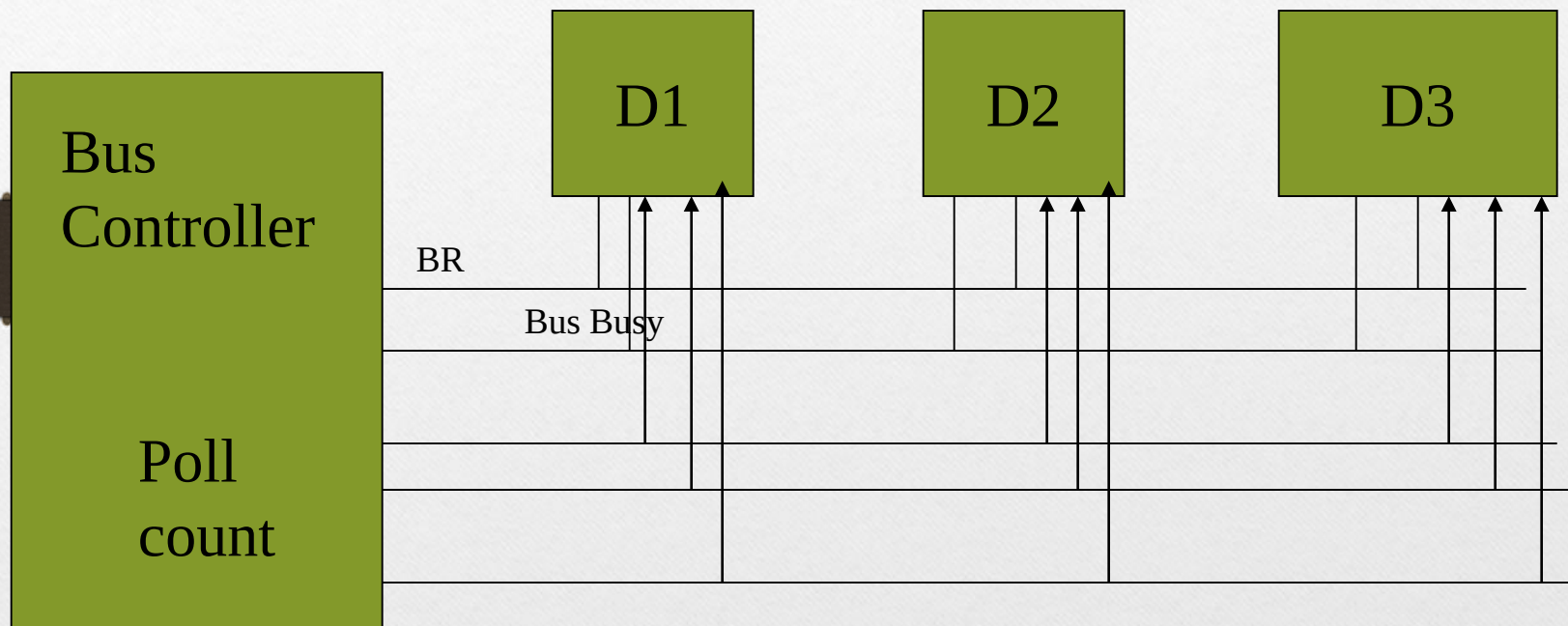


Independent requesting

- Advantage: immediate identification of all requesting units
- Disadvantage : $2n$ buses are required to control n devices



Dynamic-Polling



Polling

- Unit requests the bus via BUS request line.
- In response, the Bus controller proceeds to generate a sequence of numbers on the poll-count lines.
- Each unit compares to the unique address assigned to it.
- When requesting unit finds the match, Bus Busy signal is activated.
- In response, bus controller terminates the polling process and U_i connects to the bus.
- Advantage: failure of one unit need not affect the other units.
- Disadvantage: expensive because of more control lines. Number of units are limited based on the poll-count lines capability

Time slice:

- allocates a fixed-length time slice of bus time that is offered sequentially to each processor, in round – robin fashion.
- independent of the position along the bus.
- no preference to any particular device.

LRU/FIFO

LRU – gives highest priority to the requesting device which has not used the bus for the longest period of time.

FIFO – request are serviced in the order they receive.

Rotating Daisy chain:

- No central bus controller.
- priority –out of the last device is connected to priority in of the first device.
- current device which is accessing the bus acts as bus controller and the priority changes accordingly.